ABHIYANTRIKI 2026 · QUANT CHALLENGE · GLACIER_GAS

Surviving the flash crash with a regime-aware, risk-managed book

Data audit → chronological validation → crash detection → positions sized for Sharpe, drawdown and cost

+16.27%

our book in the crash, while the stock fell 44.9%

5.84

annualised Sharpe ratio over the crash window

-2.23%

maximum drawdown, with 16 trades in 50 days

Team only head no job

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Round 3: public 100 · private 73
Finalist, 6th of 13 teams

We audited every file before building anything

R1

Round 1: hidden rows were altered

- order_flow was computed from the open, not the close (match error 0.00005)
- volatility_20 used the open in place of the close
- rsi_14 was noisy (correlation 0.94 with the true RSI)
- Response: dropped mismatched columns; this explained our weak Round 1 score

R2

Rounds 2–3: hidden closes rebuilt exactly

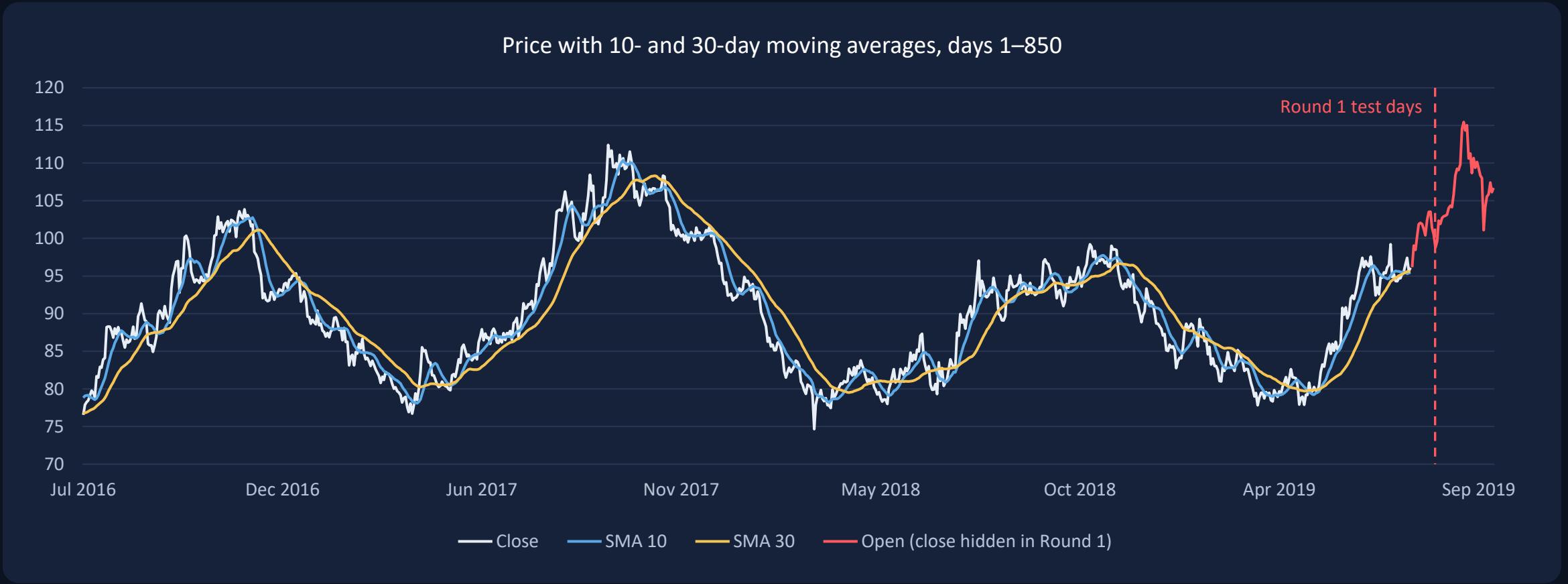
$$C_t = 10 \cdot (SMA10_t - SMA10_{t-1}) + C_{t-10}$$

- Cross-checked against RSI, volatility, order flow and momentum: max error 0.005
- Every feature now means the same thing in training and test rows

≤t

No-lookahead rule: the decision for day t uses only data up to and including day t, never row t+1. Validation is always chronological.

Slow price cycles before the test window



1

Range 75 to 112

four slow up-and-down swings in three years

2

Averages mark the turns

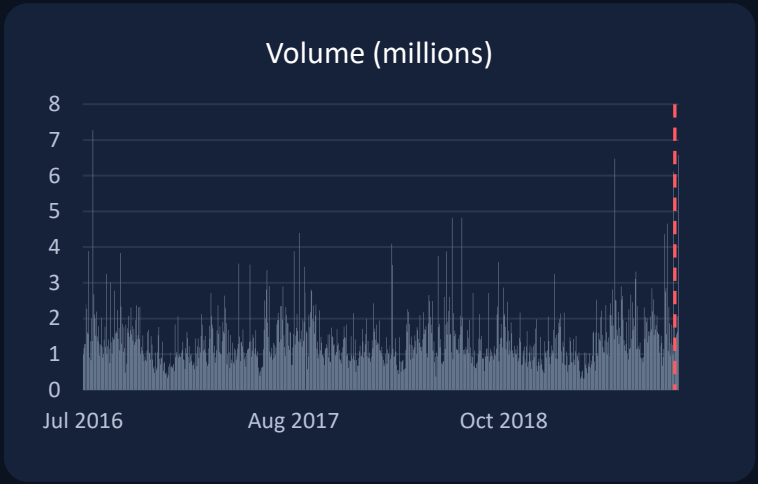
the 10-day average crossing the 30-day average flags each trend change

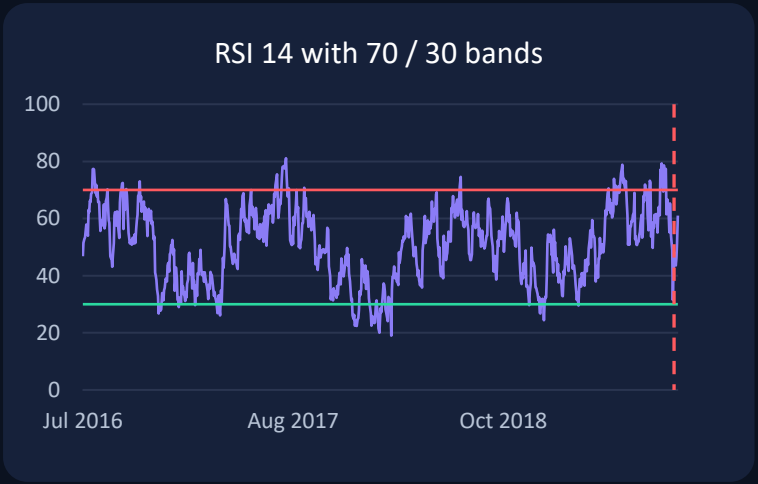
3

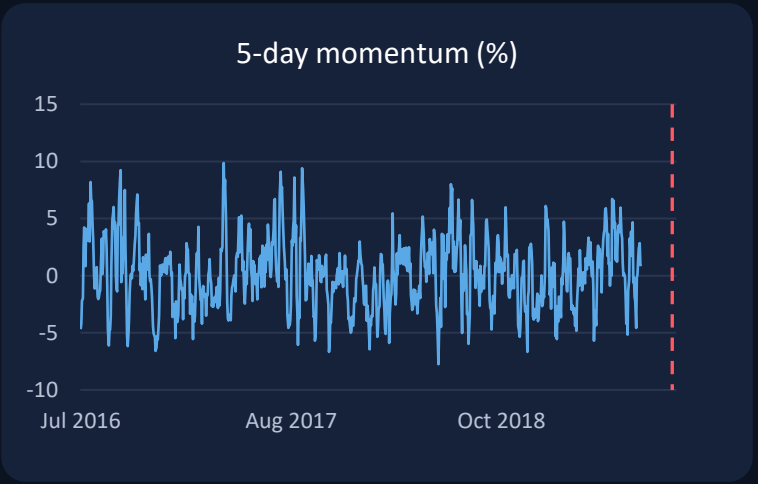
Hidden test window

days 801–850: close, returns, averages and target blank

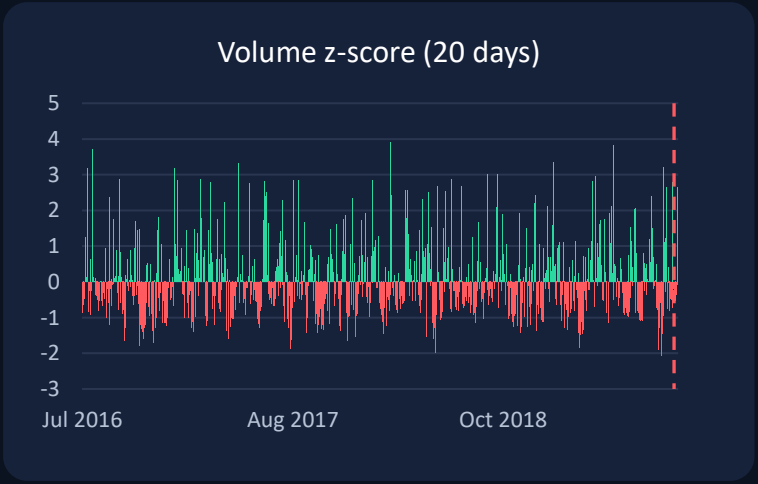
The raw indicators behind our features

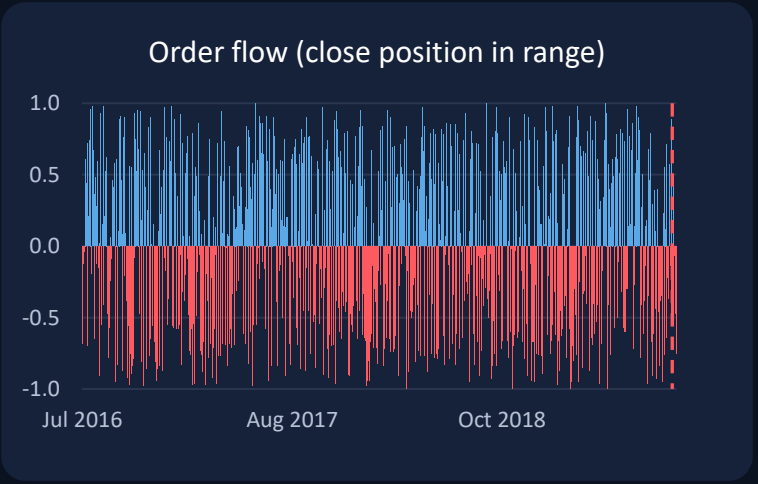












Dashed line: day 801, the start of the Round 1 hidden rows

Features encode reversal, order flow and market stress

1

Vol-scaled returns, lags 0–4

$z_t = r_t / \sigma_{20}$ keeps the same meaning in calm and crash markets

2

Order flow, lags 0–4

close location in the day's range: (2C − H − L) / (H − L)

3

RSI momentum, lags 0–4

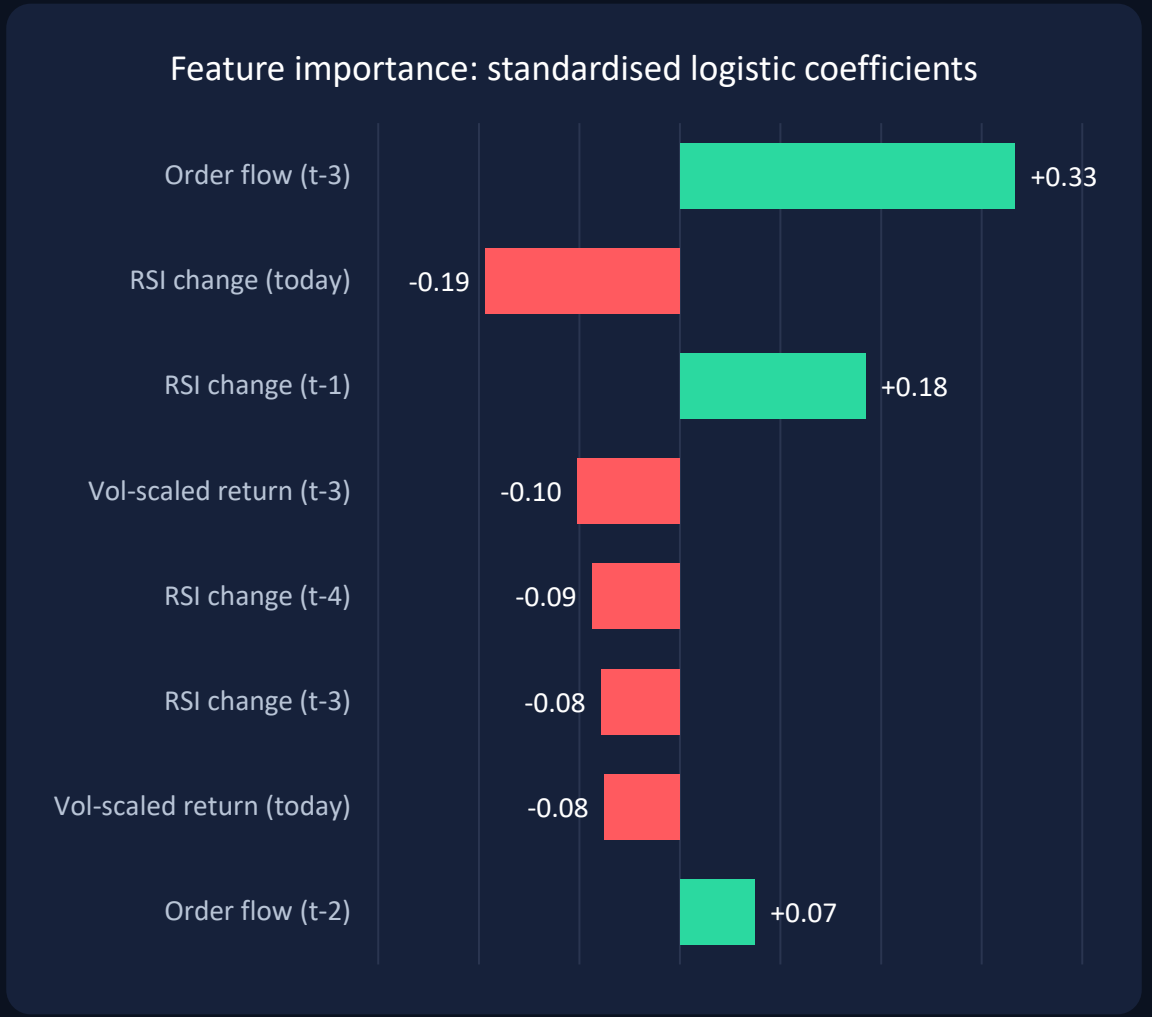
ΔRSI_{14} captures acceleration of buying or selling

4

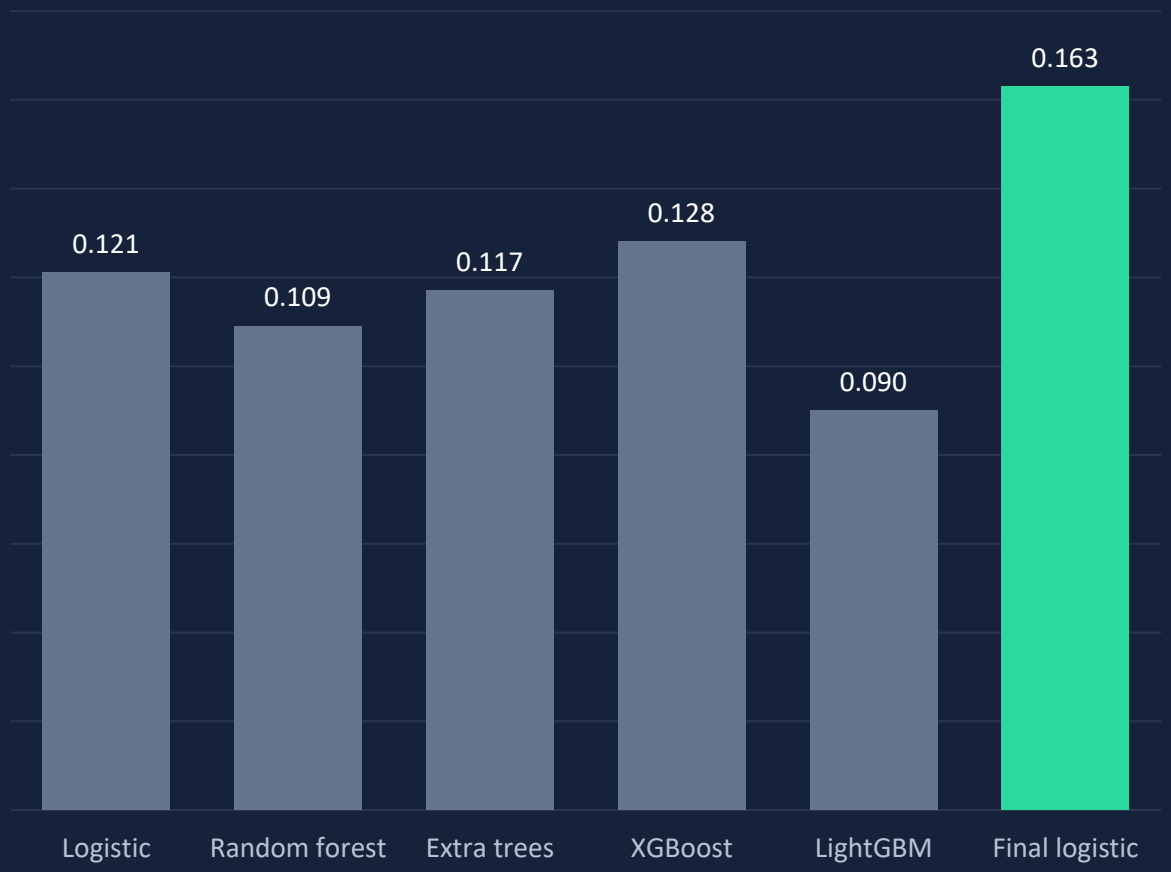
Stress and trend

drawdown $(C / \max_{20} C - 1) / \sigma_{20}$ and trend gap (SMA10 / SMA30 − 1) / σ_{20}

Why lags work: in calm markets returns alternate (today vs tomorrow −0.11, yesterday vs tomorrow +0.13)



Chronological validation chose a simple, robust model



A

Walk-forward, never shuffled

Expanding window, 1-row purge, 50-row test blocks like the real test. Scored on balanced accuracy and MCC, not raw accuracy.

B

Overfitting caught

Gradient boosting scored MCC 0.20 when selecting, then 0.02 on later untouched folds, so we rejected it.

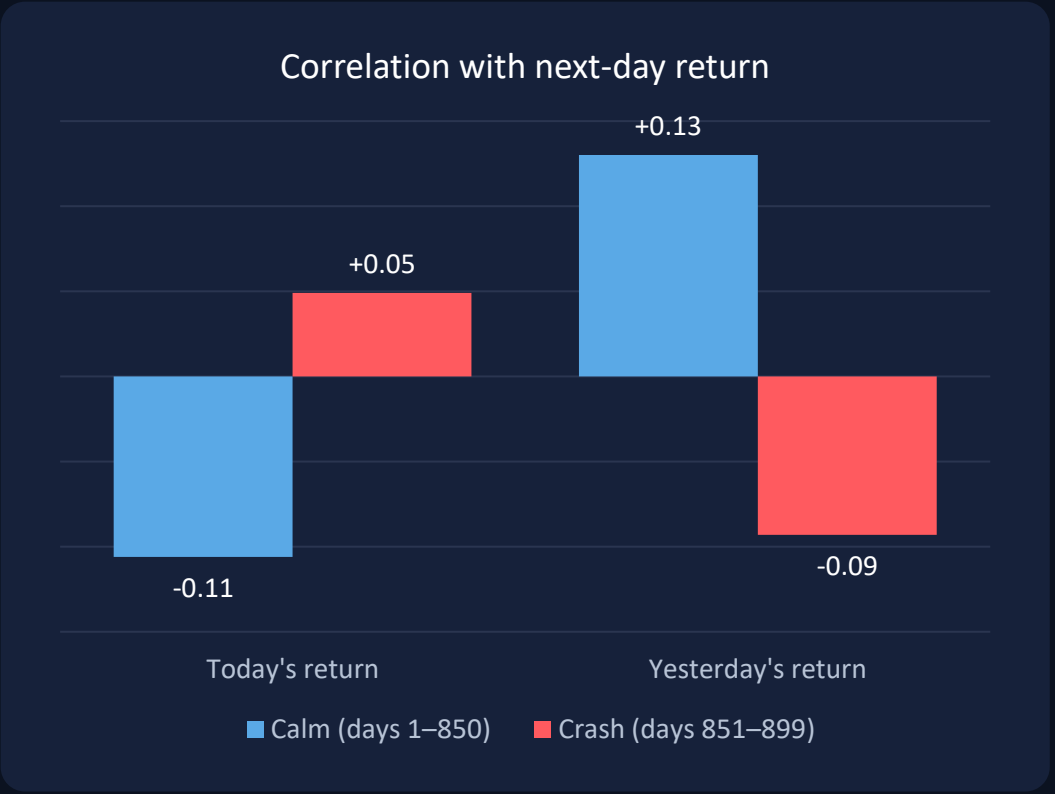
C

Final: regularised logistic ensemble

6 settings averaged. On untouched rows 801–850: MCC 0.15, balanced accuracy 0.58.

Crash: volatility doubled, market memory flipped





1 **Volatility spike**

σ_{20} above 1.3× its 250-day median; it doubled from 1.5% to 3.3%

2 **Trend break**

SMA10 crossed below SMA30 by day 850

3 **Concept drift**

reversal turned into momentum; we inverted the timing model (Round 2: 3.38 → 29.96)

Positions sized for Sharpe, drawdown and cost

1

Signal

0.7 × trend + 0.3 × timing, clipped to ±1

2

Vol targeting

× min(1, 1.2% / σ_{20}): constant daily risk

3

Drawdown guard

halve size while the book is more than 6% below its peak

4

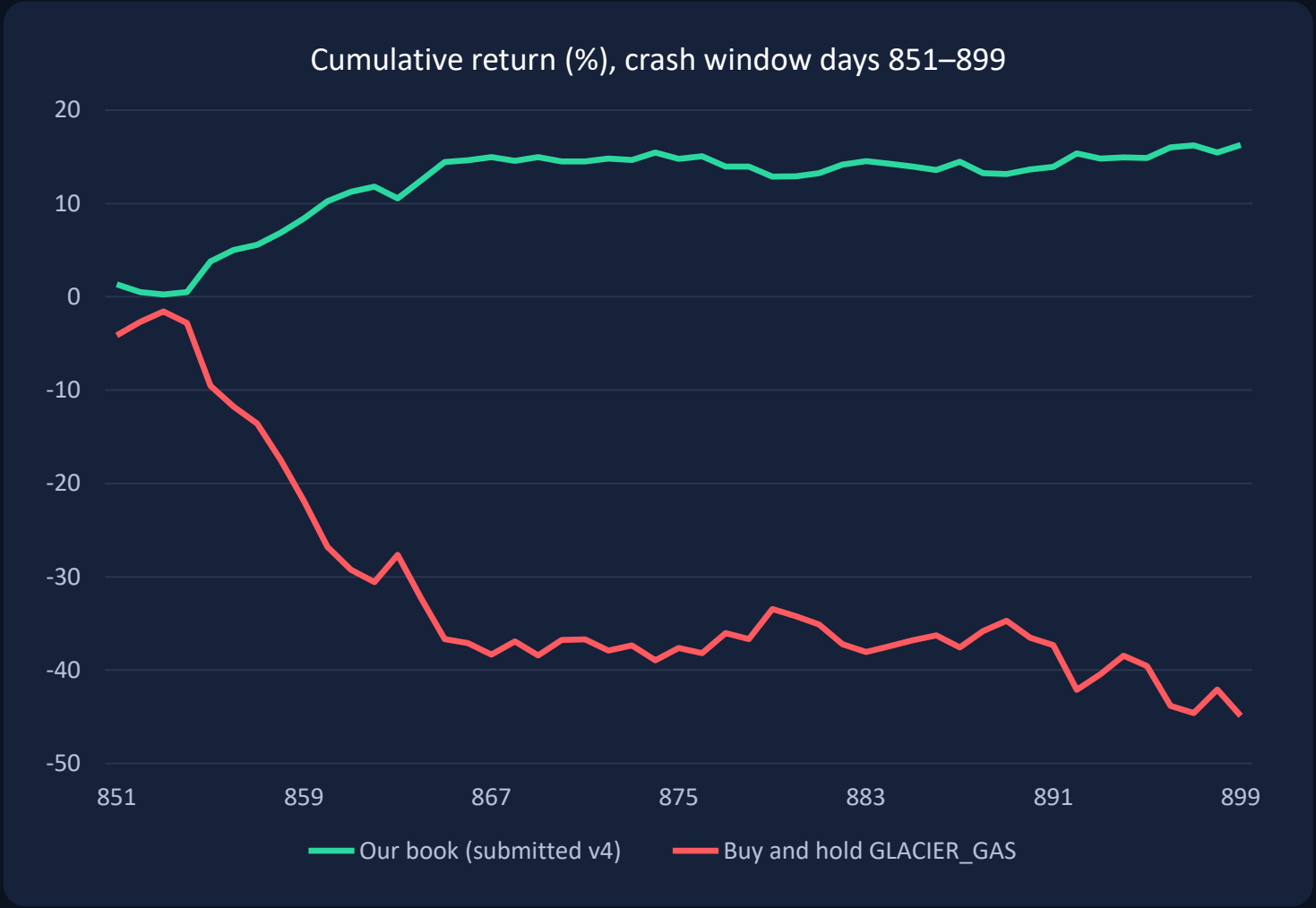
Trading band

trade only if the target moves more than 0.12 (cost 0.001 per unit)

Variant (backtest days 300–849, with costs)	Sharpe	Max drawdown	Turnover / day
Trend only	0.48	−19.6%	0.07
Previous hedged blend	0.56	−12.4%	0.49
Submitted v4 (signal + all four controls)	**0.64**	**−12.3%**	**0.13**
v4 without the drawdown guard	0.55	−17.6%	0.22

All settings were fixed on days 1–850, before the crash. The drawdown guard alone lifted Sharpe from 0.55 to 0.64 and cut the worst drawdown by a third and turnover by 40%.

Result: +16% while the stock fell 45%



5.84
annualised Sharpe ratio

-2.23%
maximum drawdown

0.332%
total trading cost (16 trades, 0.068 turnover/day)

100 / 73
Round 3 public / private score

What broke, what we fixed, what we would do next

1

Broke: Round 1 inputs

Hidden features were computed differently, so the model saw mismatched inputs.

Fix: audit every file; only use verified features.

2

Broke: calm model in the crash

The reversal pattern inverted and the unchanged model scored 3.38.

Fix: regime flags drive the model; inverted timing scored 29.96.

3

Next

Online change-point detection to switch regimes automatically.

Keep vol targeting, the drawdown guard and the band as they are.

Limitation: a 50-day test window is noisy (MCC moves about ±0.14 by chance), so we judged models on 500+ walk-forward days rather than single leaderboard scores.